

Stat 3113 Formula Sheet (Exam 2)

Continuous RV

General Properties:

- pdf: $f(x) \geq 0$, $\int_{-\infty}^{\infty} f(x)dx = 1$
- $P(a \leq X \leq b) = \int_a^b f(x)dx$
- $P(X = c) = 0$ for any specific value
- cdf: $F(x) = \int_{-\infty}^x f(t)dt$
- $P(a < X < b) = F(b) - F(a)$
- $P(X > a) = 1 - F(a)$
- $f(x) = F'(x)$

Mean & Variance:

$$\mu = \mathbb{E}[X] = \int_{-\infty}^{\infty} xf(x)dx$$

$$\sigma^2 = \text{Var}[X] = \mathbb{E}[X^2] - \mu^2$$

$$\text{where } \mathbb{E}[X^2] = \int_{-\infty}^{\infty} x^2 f(x)dx$$

Normal: $X \sim N(\mu, \sigma^2)$

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Standardize: $Z = \frac{X-\mu}{\sigma}$ where $Z \sim N(0, 1)$

68-95-99.7 Rule:

$$P(\mu - \sigma \leq X \leq \mu + \sigma) = 0.68$$

$$P(\mu - 2\sigma \leq X \leq \mu + 2\sigma) = 0.95$$

$$P(\mu - 3\sigma \leq X \leq \mu + 3\sigma) = 0.997$$

Exponential: $X \sim \text{Exp}(\lambda)$

Waiting time between events

$$f(x) = \lambda e^{-\lambda x} \text{ for } x \geq 0$$

$$F(x) = 1 - e^{-\lambda x}$$

$$\mu = \frac{1}{\lambda}, \sigma^2 = \frac{1}{\lambda^2}$$

Memoryless: $P(X > s+t | X > s) = P(X > t)$

χ^2 -Distribution

For inference on variance σ^2 .

$$\chi^2 = \frac{(n-1)s^2}{\sigma^2} \sim \chi_{n-1}^2$$

Properties: Right-skewed, df = $n - 1$, Always positive.

Sampling Distributions

Sample Mean $\bar{X}$:

$$\bullet \mathbb{E}[\bar{X}] = \mu$$

$$\bullet \text{SE}(\bar{X}) = \frac{\sigma}{\sqrt{n}}$$

Central Limit Theorem:

For $n \geq 30$, $\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$ regardless of population shape.

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1)$$

t-Distribution

Used when σ unknown.

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

df (degrees of freedom) = $n - 1$

Properties:

- Bell-shaped, symmetric at 0
- Fatter tails than normal
- As $\text{df} \rightarrow \infty$, $t \rightarrow Z$

F-Distribution

For comparing two variances.

$$F = \frac{s_1^2}{s_2^2} \text{ under } H_0 : \sigma_1^2 = \sigma_2^2$$

df: $df_1 = n_1 - 1$, $df_2 = n_2 - 1$

Properties: Right-skewed, Always positive.

Confidence Intervals

General Form: Estimate $\pm$ (Crit. Val.) $\times$ SE

One-Sample Mean (σ known):

$$\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

One-Sample Mean (σ unknown):

$$\bar{x} \pm t_{\alpha/2, n-1} \frac{s}{\sqrt{n}}, \text{ df} = n - 1$$

One-Sample Proportion:

$$\hat{p} \pm z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

Two Independent Samples (Means):

$$(\bar{x}_1 - \bar{x}_2) \pm t_{\alpha/2, \text{df}} \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

Paired Samples (Mean Diff):

$$\bar{d} \pm t_{\alpha/2, n-1} \frac{s_d}{\sqrt{n}}, \text{ df} = n - 1$$

Common Critical Values (z):

- 90% CI: $z_{0.05} = 1.645$
- 95% CI: $z_{0.025} = 1.96$
- 99% CI: $z_{0.005} = 2.576$

Hypothesis Testing

General Test Stat: $\frac{\text{Est.} - \text{Null Val.}}{\text{SE}}$

Decision Rule: p-val $\leq \alpha \implies$ Reject H_0

One-Sample Mean (σ known):

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$$

One-Sample Mean (σ unknown):

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}, \text{ df} = n - 1$$

One-Sample Proportion: $z =$

$$\frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

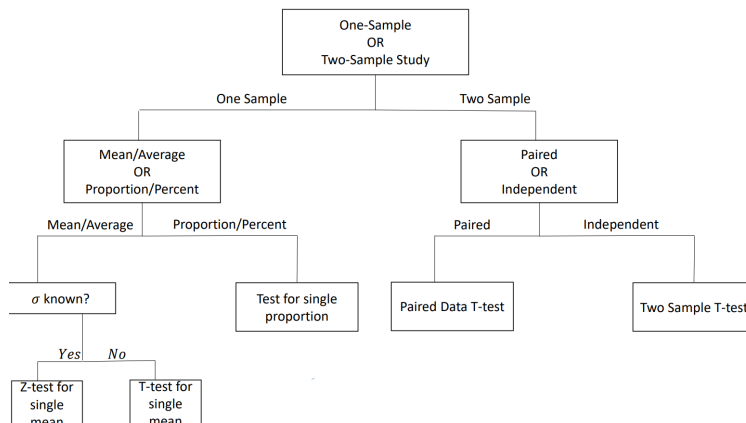
Two Independent Samples: $t =$

$$\frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}, \text{ Use Welch's df}$$

Paired Samples: $t = \frac{\bar{d} - 0}{s_d/\sqrt{n}}, \text{ df} = n - 1$

Errors:

- **Type I:** Reject true H_0 (prob = α)
- **Type II:** Fail to reject false H_0 (prob = β)
- **Power:** $1 - \beta$ (Prob. of detecting a true effect)



Stat 3113 Definitions & Concepts (Exam 2)

Continuous Distributions

Continuous Random Variable: Takes any value in an interval

Probability Density Function (pdf): Describes likelihood of values Area under curve = probability

Cumulative Distribution Function (cdf): $F(x) = P(X \leq x)$ Always non-decreasing

Key Property: $P(X = c) = 0$ for any specific value

Normal Distribution

Properties:

- Bell-shaped, symmetric
- Mean = median = mode
- Completely determined by μ and σ
- Total area under curve = 1

Standard Normal (Z):

- $\mu = 0, \sigma = 1$
- Used for probability tables
- Standardize: $Z = \frac{X - \mu}{\sigma}$

Empirical Rule:

- 68% within 1 std dev
- 95% within 2 std devs
- 99.7% within 3 std devs

Exponential Distribution

Use: Models waiting times

Parameter λ : Rate of occurrence

Memoryless Property: Past doesn't affect future waiting time "Starting over" at any point

Relationship to Poisson: If events follow Poisson(λ), time between events is Exp(λ)

Sampling Concepts

Sampling Distribution: Distribution of a sample statistic

Standard Error (SE): Standard deviation of sampling distribution Measures precision of estimate

Point Estimate: Single best guess for parameter $\bar{x}$ estimates μ , $\hat{p}$ estimates p

Unbiased Estimator: Expected value equals parameter $\mathbb{E}[\bar{X}] = \mu$, $\mathbb{E}[\hat{p}] = p$

Central Limit Theorem

Key Idea: For large samples ($n \geq 30$), $\bar{X}$ is approximately normal regardless of population shape

Implications:

- Can use normal-based inference
- Larger n = better approximation
- Reduces influence of outliers

T-Distribution

When to Use: Population σ unknown Estimate with sample s

Degrees of Freedom (df): Amount of independent information Usually $df = n - 1$

Comparison to Normal:

- More spread (fatter tails)
- Accounts for uncertainty in s
- Approaches normal as df increases
- More conservative (wider CIs)

Chi-Squared Distribution

Use: Inference for variance σ^2

Properties:

- Right-skewed
- Only positive values
- Shape depends on df
- Less skewed as df increases

Application: Test if population variance equals hypothesized value

F-Distribution

Use: Compare two variances

Parameters:

Two df values: df_1 and df_2

From two samples

Properties:

- Right-skewed
- Not symmetric
- $F \geq 0$ always

Application: ANOVA, regression (later in course)

Confidence Intervals

Interpretation: We are C% confident the interval contains the true parameter

Confidence Level: Probability method works

Most common: 90%, 95%, 99%

Margin of Error:

(Critical Value) $\times$ SE

Half the width of CI

Factors Affecting Width:

- Sample size n (larger n = narrower)
- Confidence level (higher = wider)
- Variability (higher s = wider)

Two-Sample Inference

Independent Samples:

Two separate groups
No pairing or matching

Paired Samples:

Natural pairing exists
Before/after, matched pairs
Analyze differences d_i

Welch's t-test:

Doesn't assume equal variances
More conservative df calculation
Preferred in practice

Hypothesis Testing

Null Hypothesis (H_0):

Status quo, no effect
What we assume true initially

Alternative Hypothesis (H_a):

Research claim
What we want evidence for

p-value:

Probability of observing data this extreme if H_0 is true
Small p-value = evidence against H_0

Significance Level (α):

Threshold for decision
Common: 0.05, 0.01, 0.10

Statistically Significant:

p-value $< \alpha$
Reject H_0

Practical Significance:

Effect size matters, not just p-value
Large sample = small p-values easier

Test Conclusions

Reject H_0 : Sufficient evidence for H_a

Fail to Reject H_0 :

Insufficient evidence for H_a
NOT the same as "accepting H_0 "

Never say:

- "Prove" or "accept"
- Only "evidence for/against"

Type I & II Errors

Type I Error (α):

False positive
Reject true H_0
Probability controlled by α

Type II Error (β):

False negative
Fail to reject false H_0
Affected by effect size, n , α

Power ($1 - \beta$):

Probability of detecting true effect
Correctly reject a false null hypothesis
Higher power is better

Trade-off:

Decreasing α increases β
Increase n to reduce both